

Answer all questions.

1. The images below show a longitudinal section through a tissue found lining the trachea of humans and a highly magnified image of the surface of this tissue.

Image 1

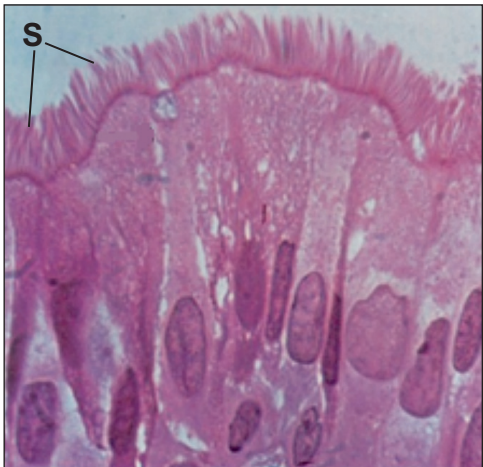
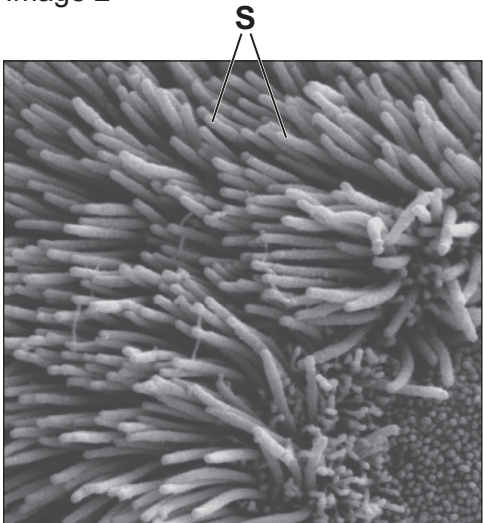


Image 2



- (a) (i) Name the structures labelled **S** and suggest their function in the human respiratory system. [2]

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- (ii) Explain why Image 1 is referred to as a tissue rather than an organ. [1]

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The photo electron micrograph below shows two mitochondria found in this tissue.



- (b) (i) One reason for the different appearance of the mitochondria is that they have been cut in different planes. Suggest **one other** reason that could account for the difference in their appearance. [1]

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- (ii) Calculate the actual size of mitochondrion **T** along the line shown on the image. Give your answer in micrometres. [3]

Length of organelle = μm

- (iii) State and explain **one** way in which mitochondria are adapted to perform their function. [2]

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